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《高等数值计算方法》

第四章：数值积分和数值微分

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4.6 数值微分

知识结构图

数值微分
- 中点方法（Taylor展开型方法）
- 插值型求导公式

4.6 数值微分

4.6.1 中点方法与误差分析

如果函数$f(x)$以离散点列给出，或函数$f(x)$过于复杂，而要求我们给出导数值时，这两种情况都要求我们用数值的方法求函数的导数值。

数值微分就是要用函数值的线性组合近似函数在某点的导数值。由导数定义差商来近似导数，得到数值微分公式。

微积分中，关于导数的定义如下：

$$f'(x)=\lim_{h\to 0}\frac{f(x+h)-f(x)}{h}=\lim_{h\to 0}\frac{f(x)-f(x-h)}{h}=\lim_{h\to 0}\frac{f(x+h)-f(x-h)}{2h}$$

自然，而又简单的方法就是，取极限的近似值，即差商。

4.6.1 中点方法与误差分析

差商型求导公式

由导数定义$f'(x)=\lim\limits_{h\to 0}\dfrac{f(x+h)-f(x)}{h}$

- 向前差商公式

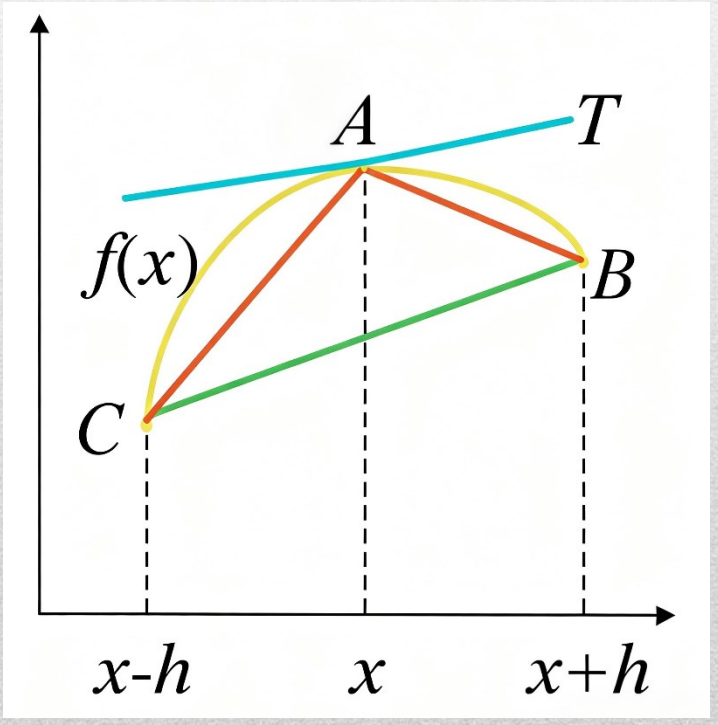
$$f'(x)\approx\frac{f(x+h)-f(x)}{h}$$

- 向后差商公式

$$f'(x)\approx\frac{f(x)-f(x-h)}{h}$$

- 中心差商公式（中点方法）

$$f'(x)\approx\frac{f(x+\mathrm{h})-f(x-h)}{2h}$$



4.6.1 中点方法与误差分析

差商型数值微分

（1）**向前差商数值微分公式**

$$f'(x_0) \approx \frac{f(x_0+h)-f(x_0)}{h}$$

由Taylor展开式

$$f(x_0+h) = f(x_0) + f'(x_0)h + \frac{h^2}{2}f''(x_0+\theta h),\ 0 \le \theta \le 1$$

可得误差

$$f'(x_0) - \frac{f(x_0+h)-f(x_0)}{h} = -\frac{h}{2}f''(x_0+\theta h),\ 0 \le \theta \le 1$$

拉格朗日（Lagrange）余项形式

4.6.1 中点方法与误差分析

- **均匀网格下的截断误差**

$$\begin{cases} f(x_{i+1}) = f(x_i + h) = f(x_i) + f'(x_i)h + \dfrac{h^2}{2!}f''(x_i) + \dfrac{h^3}{3!}f'''(x_i) + \cdots \\ f(x_{i-1}) = f(x_i - h) = f(x_i) - f'(x_i)h + \dfrac{h^2}{2!}f''(x_i) - \dfrac{h^3}{3!}f'''(x_i) + \cdots \end{cases}$$

$$\begin{cases} \text{Forward:} \quad f'(x_i) = \dfrac{f(x_{i+1}) - f(x_i)}{h} - \dfrac{h}{2}f''(\xi_1) & O(h) \\ \text{Backward:} \quad f'(x_i) = \dfrac{f(x_i) - f(x_{i-1})}{h} + \dfrac{h}{2}f''(\xi_2) & O(h) \\ \text{Central:} \quad f'(x_i) = \dfrac{f(x_{i+1}) - f(x_{i-1})}{2h} - \dfrac{h^2}{6}f'''(\xi_3) & O(h^2) \end{cases}$$

4.6.1 中点方法与误差分析

(2) **中心差商数值微分公式**

$$f'(x_0) \approx \frac{f(x_0+h)-f(x_0-\mathrm{h})}{2h}$$

由Taylor展开式

$$f(x_0+h) = f(x_0) + f'(x_0)h + \frac{h^2}{2!}f''(x_0) + \frac{h^3}{3!}f'''(\xi_1),\quad x_0 \le \xi_1 \le x_0+h$$

$$f(x_0-h) = f(x_0) - f'(x_0)h + \frac{h^2}{2!}f''(x_0) + \frac{h^3}{3!}f'''(\xi_2),\quad x_0-h \le \xi_2 \le x_0$$

因此，有误差

$$R(x) = f'(x_0) - \frac{f(x_0+h)-f(x_0-h)}{2h}$$

$$= \frac{h^2}{12}\left[f'''(\xi_1)+f'''(\xi_2)\right] = \frac{h^2}{6}f'''(\xi) = O(h^2)$$

介值定理

Example: First Derivatives

- **Use forward and backward difference approximations to estimate the first derivative of**

$$f(x) = -0.1x^4 - 0.15x^3 - 0.5x^2 - 0.25x + 1.2$$

at $x = 0.5$ with $h = 0.5$ and 0.25 (exact solution = -0.9125)

- **Forward Difference**

$$\begin{cases} h = 0.5, \ f'(0.5) \approx \dfrac{f(1) - f(0.5)}{1 - 0.5} = \dfrac{0.2 - 0.925}{0.5} = -1.45, & |\varepsilon_t| = 58.9\% \\ h = 0.25, \ f'(0.5) \approx \dfrac{f(0.75) - f(0.5)}{0.75 - 0.5} = \dfrac{0.63632813 - 0.925}{0.25} = -1.155, & |\varepsilon_t| = 26.5\% \end{cases}$$

- **Backward Difference**

$$\begin{cases} h = 0.5, \ f'(0.5) \approx \dfrac{f(0.5) - f(0)}{0.5 - 0} = \dfrac{0.925 - 1.2}{0.5} = -0.55, & |\varepsilon_t| = 39.7\% \\ h = 0.25, \ f'(0.5) \approx \dfrac{f(0.5) - f(0.25)}{0.5 - 0.25} = \dfrac{0.925 - 1.10351563}{0.25} = -0.714, & |\varepsilon_t| = 21.7\% \end{cases}$$

Example: First Derivatives

- **Use forward and backward difference approximations to estimate the first derivative of**

$$f(x) = -0.1x^4 - 0.15x^3 - 0.5x^2 - 0.25x + 1.2$$

at $x = 0.5$ with $h = 0.5$ and 0.25 (exact solution = -0.9125)

- **Central Difference**

$$h = 0.5, \quad f'(0.5) \approx \frac{f(1) - f(0)}{1 - 0} = \frac{0.2 - 1.2}{1} = -1.0, \qquad |\varepsilon_t| = 9.6\%$$

$$h = 0.25, \quad f'(0.5) \approx \frac{f(0.75) - f(0.25)}{0.75 - 0.25}$$

$$= \frac{0.63632813 - 1.10351563}{0.5} = -0.934, \quad |\varepsilon_t| = 2.4\%$$

4.6.1 中点方法与误差分析

- **均匀网格下的舍入误差**

$$f(a\pm h)=f(a)\pm f'(a)h+\frac{h^2}{2!}f''(a)\pm\frac{h^3}{3!}f'''(a)+\frac{h^4}{4!}f^{(4)}(a)\pm\frac{h^5}{5!}f^{(5)}(a)+\cdots$$

$$G(h)\triangleq\frac{f(a+h)-f(a-h)}{2h}=f'(a)+\frac{h^3}{3!}f'''(a)+\frac{h^5}{5!}f^{(5)}(a)+\cdots$$

误差估计$|G(h)-f'(a)|\le\frac{h^2}{6}M$，其中$M\ge\max\limits_{|x-a|\le h}|f'''(x)|$

表面上看h越小越好，但从舍入误差角度考虑，h不能太小。

例：$f(x)=\sqrt{x}$，在$x=2$处的一阶导数 $\left(\frac{1}{2}*\frac{1}{\sqrt{2}}=0.35353390593274\right)$

$G(h)=\frac{\sqrt{2+h}-\sqrt{2+h}}{2h}$，设取4位数字计算。

4.6.1 中点方法与误差分析

h	$G(h)$	h	$G(h)$	h	$G(h)$
1	0.3660	0.05	0.3530	0.001	0.3500
0.5	0.3564	0.01	0.3500	0.0005	0.3000
0.1	0.3535	0.005	0.3500	0.0001	0.3000

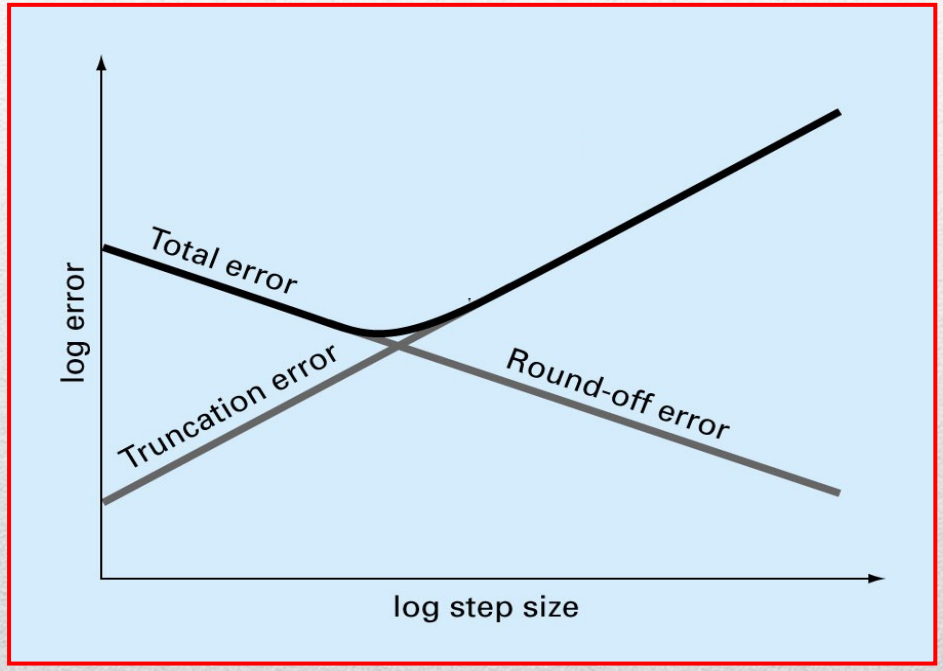
从表中看到$h = 0.1$的逼近效果最好，如果进一步缩小步长，则逼近效果反而有差。这是因为当$f(a+h)$及$f(a-h)$分别有舍入误差ϵ_1及ϵ_2。令$\epsilon = \max\{|\epsilon_1|, |\epsilon_2|\}$，则计算的舍入误差上界为

$$\delta(f'(a)) = |f'(a) - G(a)| \leq \frac{|\epsilon_1|+|\epsilon_2|}{2h} \leq \frac{\epsilon}{h}$$

表明h越小，舍入误差越大，故它是病态的。

4.6.1 中点方法与误差分析

- 均匀网格下的数值计算总误差



Trade-off between truncation and round-off errors

4.6.1 中点方法与误差分析

例： **Round-off errors**

Table 6.1 Finding the Difference Quotients $D_k = (e^{1+h_k} - e)/h_k$

h_k	$f_k = f(1+h_k)$	$f_k - e$	$D_k = (f_k - e)/h_k$
$h_1 = 0.1$	3.004166024	0.285884196	2.858841960
$h_2 = 0.01$	2.745601015	0.027319187	2.731918700
$h_3 = 0.001$	2.721001470	0.002719642	2.719642000
$h_4 = 0.0001$	2.718553670	0.000271842	2.718420000
$h_5 = 0.00001$	2.718309011	0.000027183	2.718300000
$h_6 = 10^{-6}$	2.718284547	0.000002719	2.719000000
$h_7 = 10^{-7}$	2.718282100	0.000000272	2.720000000
$h_8 = 10^{-8}$	2.718281856	0.000000028	2.800000000
$h_9 = 10^{-9}$	2.718281831	0.000000003	3.000000000
$h_{10} = 10^{-10}$	2.718281828	0.000000000	0.000000000

4.6.2 插值型的求导公式

已知函数$y=f(x)$的节点上的函数值$y_i=f(x_i)(i=0,1,\cdots,n)$，建立插值多项式$P(x)$

取 $$f'(x)\approx P'(x),$$

统称为插值型求导公式。

函数插值误差余项为$f(x)-P_n(x)=\frac{f^{(n+1)}(\xi)}{(n+1)!}\omega_{n+1}(x)$

导数误差余项为

$$f'(x)-P_n'(x)=\frac{f^{(n+1)}(\xi)}{(n+1)!}\omega_{n+1}'(x)+\frac{\omega_{n+1}(x)}{(n+1)!}\frac{\mathrm{d}}{\mathrm{d}x}f^{(n+1)}(\xi),$$

其中$\xi\in(a,b),\omega_{n+1}(x)=\prod_{j=0}^{n}(x-x_j)$

$$\Rightarrow f'(x_k)-P_n'(x_k)=\frac{f^{(n+1)}(\xi)}{(n+1)!}\omega_{n+1}'(x_k)$$

4.6.2 插值型的求导公式

插值型求导公式

利用插值原理，建立插值多项式$y = L_n(x)$作为$f(x)$的近似。由于多项式求导比较容易，取

$$f'(\mathrm{x}) \approx L_n{}'(x)$$

但是，需要注意误差分析

$$R_n(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!}\omega_n(x) = f(x) - L_n(x)$$

$$f'(x) - L_n'(x) = \frac{d}{dx}\left[\frac{f^{(n+1)}(\xi)}{(n+1)!}\omega_{n+1}(x)\right]$$

$$= \frac{d}{dx}\left[\frac{f^{(n+1)}(\xi)}{(n+1)!}\right]\omega_{n+1}(x) + \frac{f^{(n+1)}(\xi)}{(n+1)!}\frac{d}{dx}[\omega_{n+1}(x)]$$

ξ是x的函数，此项无法估计

4.6.2 插值型的求导公式

所以，对于任意给出的点x，误差无法预估的。但是，如果限定求某个节点处的导数值时，上述第一项为零，这时余项公式为

$$\frac{f^{(n+1)}(\xi)}{(n+1)!}[\omega_{n+1}{}'(x)]$$

4.6.2 插值型的求导公式

下面考虑在等距节点时，节点上的导数值，两点公式

$$P_1(x)=\frac{x-x_1}{x_0-x_1}f(x_0)+\frac{x-x_0}{x_1-x_0}f(x_1),$$

$$P_1'(x)=\frac{1}{h}\left[-f(x_0)+f(x_1)\right],$$

$$P_1'(x_0)=\frac{1}{h}\left[f(x_1)-f(x_0)\right],\quad P_1'(x_1)=\frac{1}{h}\left[f(x_1)-f(x_0)\right].$$

$$f'(x_0)=\frac{1}{h}\left[f(x_1)-f(x_0)\right]-\frac{h}{2}f''(\xi),$$

$$f'(x_1)=\frac{1}{h}\left[f(x_1)-f(x_0)\right]+\frac{h}{2}f''(\xi).$$

4.6.2 插值型的求导公式

取$n=2$的等距节点$(x_2-x_1=x_1-x_0=h)$抛物线插值:

$$L_2(x)=[(x-x_1)(x-x_2)f(x_0)-2(x-x_0)(x-x_2)f(x_1)+(x-x_0)(x-x_1)f(x_2)]/2h^2$$

则有

$$L_2'(x)=[(2x-x_1-x_2)f(x_0)-2(2x-x_0-x_2)f(x_1)+(2x-x_0-x_1)f(x_2)]/2h^2$$

$$L_2''(x)=[2f(x_0)-4f(x_1)+2f(x_2)]/2h^2$$

可得数值微分的三点公式:

$$\begin{cases} f'(x_0)=\dfrac{1}{2h}[-3f(x_0)+4f(x_1)-f(x_2)]+\dfrac{h^2}{3}f'''(\xi) \\ f'(x_1)=\dfrac{1}{2h}[-f(x_0)+f(x_2)]-\dfrac{h^2}{6}f'''(\xi) \\ f'(x_2)=\dfrac{1}{2h}[f(x_0)-4f(x_1)+3f(x_2)]+\dfrac{h^2}{3}f'''(\xi) \\ f''(x_0)=\dfrac{1}{h^2}[f(x_0)-2f(x_1)+f(x_2)]+[-hf'''(\xi_1)+\dfrac{h^2}{6}f^{(+)}(\xi_2)] \\ f''(x_1)=\dfrac{1}{h^2}[f(x_0)-2f(x_1)+f(x_2)]-\dfrac{h^2}{12}f^{(+)}(\xi) \\ f''(x_2)=\dfrac{1}{h^2}[f(x_0)-2f(x_1)+f(x_2)]+[hf'''(\xi_1)-\dfrac{h^2}{6}f^{(+)}(\xi_2)] \end{cases}$$

误差次项系数对否？

REPORT：7/12

General Three-Point Formula

- **Lagrange interpolation polynomial for unequally spaced data**

$$f(x) \approx l_{i-1}(x) f(x_{i-1}) + l_i(x) f(x_i) + l_{i+1}(x) f(x_{i+1})$$
$$= f(x_{i-1}) \frac{(x - x_i)(x - x_{i+1})}{(x_{i-1} - x_i)(x_{i-1} - x_{i+1})} + f(x_i) \frac{(x - x_{i-1})(x - x_{i+1})}{(x_i - x_{i-1})(x_i - x_{i+1})}$$
$$+ f(x_{i+1}) \frac{(x - x_{i-1})(x - x_i)}{(x_{i+1} - x_{i-1})(x_{i+1} - x_i)}$$

- **First derivative**

$$f'(x) \approx f(x_{i-1}) \frac{2x - x_i - x_{i+1}}{(x_{i-1} - x_i)(x_{i-1} - x_{i+1})} + f(x_i) \frac{2x - x_{i-1} - x_{i+1}}{(x_i - x_{i-1})(x_i - x_{i+1})}$$
$$+ f(x_{i+1}) \frac{2x - x_{i-1} - x_i}{(x_{i+1} - x_{i-1})(x_{i+1} - x_i)}$$

Three-point formula (Uniform Grid)

$$f'(x) \approx f(x_{i-1})\frac{2x-x_i-x_{i+1}}{2h^2} + f(x_i)\frac{2x-x_{i-1}-x_{i+1}}{-h^2} + f(x_{i+1})\frac{2x-x_{i-1}-x_i}{2h^2}$$

- **Forward difference**

$$f'(x_{i-1}) \approx f(x_{i-1})\frac{2x_{i-1}-x_i-x_{i+1}}{2h^2} + f(x_i)\frac{2x_{i-1}-x_{i-1}-x_{i+1}}{-h^2} + f(x_{i+1})\frac{2x_{i-1}-x_{i-1}-x_i}{2h^2}$$
$$= \frac{-3h}{2h^2}f(x_{i-1}) + \frac{-2h}{-h^2}f(x_i) + \frac{-h}{2h^2}f(x_{i+1}) = \frac{-3f(x_{i-1})+4f(x_i)-f(x_{i+1})}{2h}$$

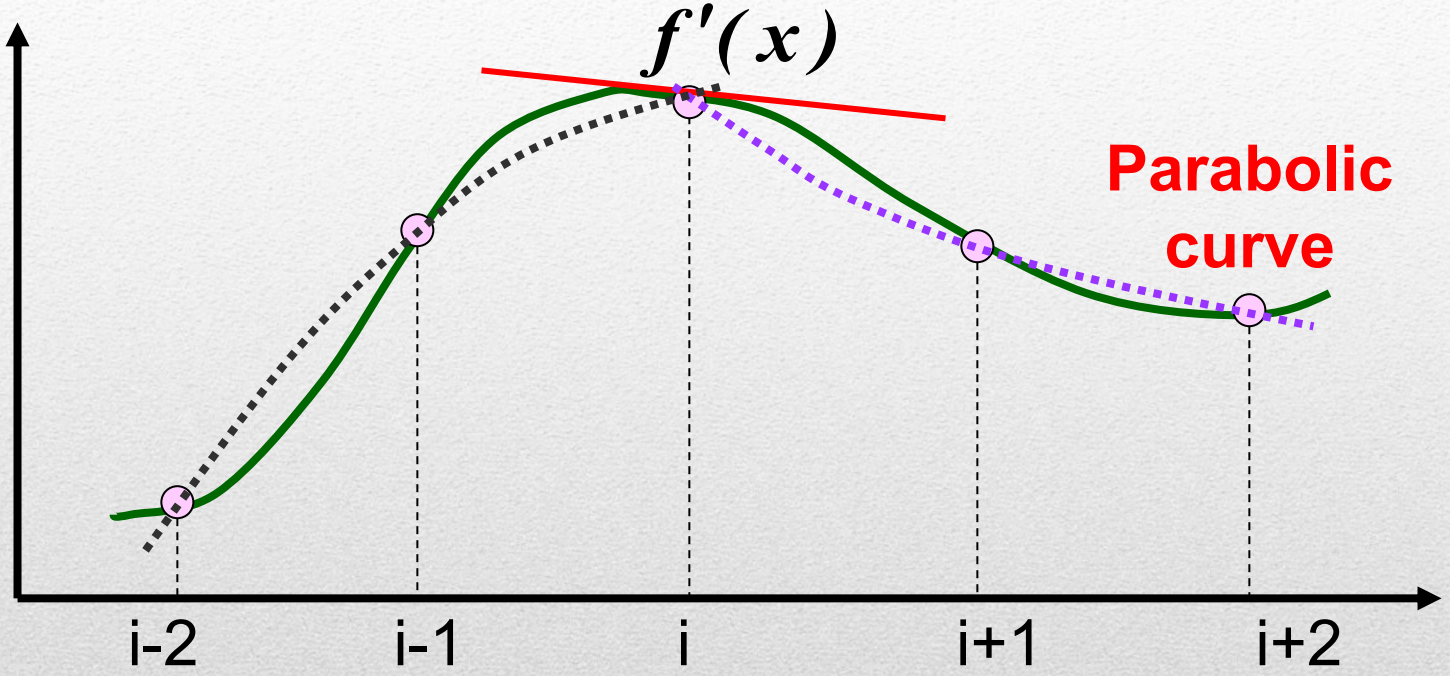
- **Centered difference**

$$f'(x_i) \approx f(x_{i-1})\frac{2x_i-x_i-x_{i+1}}{2h^2} + f(x_i)\frac{2x_i-x_{i-1}-x_{i+1}}{-h^2} + f(x_{i+1})\frac{2x_i-x_{i-1}-x_i}{2h^2}$$
$$= \frac{-h}{2h^2}f(x_{i-1}) + \frac{h}{2h^2}f(x_{i+1}) = \frac{f(x_{i+1})-f(x_{i-1})}{2h}$$

- **Backward difference**

$$f'(x_{i+1}) \approx f(x_{i-1})\frac{2x_3-x_2-x_3}{2h^2} + f(x_i)\frac{2x_3-x_1-x_3}{-h^2} + f(x_{i+1})\frac{2x_3-x_1-x_2}{2h^2}$$
$$= \frac{h}{2h^2}f(x_{i-1}) + \frac{2h}{-h^2}f(x_i) + \frac{3h}{2h^2}f(x_{i+1}) = \frac{f(x_{i-1})-4f(x_i)+3f(x_{i+1})}{2h}$$

First Derivatives



- **3 -point Forward difference** $f'(x_i) \approx \dfrac{-f(x_{i+2}) + 4f(x_{i+1}) - 3f(x_i)}{x_{i+2} - x_i}$

- **3 -point Backward difference** $f'(x_i) \approx \dfrac{3f(x_i) - 4f(x_{i-1}) + f(x_{i-2})}{x_i - x_{i-2}}$

Example: First Derivatives

- **Use forward and backward difference approximations to estimate the first derivative of**

$$f(x) = -0.1x^4 - 0.15x^3 - 0.5x^2 - 0.25x + 1.2$$

at $x = 0.5$ with $h = 0.5$ and 0.25 (exact solution = -0.9125)

- **Forward Difference**

$$f'(0.5) \approx \frac{-f(1) + 4f(0.75) - 3f(0.5)}{2(0.25)}$$

$$= \frac{-0.2 + 4(0.6363281) - 3(0.925)}{0.5} = -0.859375, \quad |\varepsilon_t| = 5.82\%$$

- **Backward Difference**

$$f'(0.5) \approx \frac{3f(0.5) - 4f(0.25) + f(0)}{2(0.25)}$$

$$= \frac{3(0.925) - 4(1.035156) + 1.2}{0.5} = -0.878125, \quad |\varepsilon_t| = 3.77\%$$



Centered Finite-Divided Differences

First Derivative

Error

$$f'(x_i) = \frac{f(x_{i+1}) - f(x_{i-1})}{2h} \qquad O(h^2)$$

$$f'(x_i) = \frac{-f(x_{i+2}) + 8f(x_{i+1}) - 8f(x_{i-1}) + f(x_{i-2})}{12h} \qquad O(h^4)$$

Second Derivative

$$f''(x_i) = \frac{f(x_{i+1}) - 2f(x_i) + f(x_{i-1})}{h^2} \qquad O(h^2)$$

$$f''(x_i) = \frac{-f(x_{i+2}) + 16f(x_{i+1}) - 30f(x_i) + 16f(x_{i-1}) - f(x_{i-2})}{12h^2}$$

$$O(h^4)$$

Centered Finite-Divided Differences

Third Derivative

Error

$$f'''(x_i) = \frac{f(x_{i+2}) - 2f(x_{i+1}) + 2f(x_{i-1}) - f(x_{i-2})}{2h^3} \qquad O(h^2)$$

$$f'''(x_i) = \frac{-f(x_{i+3}) + 8f(x_{i+2}) - 13f(x_{i+1}) + 13f(x_{i-1}) - 8f(x_{i-2}) + f(x_{i-3})}{8h^3} \qquad O(h^4)$$

Fourth Derivative

$$f^{(4)}(x_i) = \frac{f(x_{i+2}) - 4f(x_{i+1}) + 6f(x_i) - 4f(x_{i-1}) + f(x_{i-2})}{h^4} \qquad O(h^2)$$

$$f^{(4)}(x_i) = \frac{-f(x_{i+3}) + 12f(x_{i+2}) + 39f(x_{i+1}) + 56f(x_i) - 39f(x_{i-1}) + 12f(x_{i-2}) + f(x_{i-3})}{6h^4}$$

$$O(h^4)$$



Forward Finite-Divided Differences

First Derivative **Error**

$$f'(x_i) = \frac{f(x_{i+1}) - f(x_i)}{h} \qquad O(h)$$

$$f'(x_i) = \frac{-f(x_{i+2}) + 4f(x_{i+1}) - 3f(x_i)}{2h} \qquad O(h^2)$$

Second Derivative

$$f''(x_i) = \frac{f(x_{i+2}) - 2f(x_{i+1}) + f(x_i)}{h^2} \qquad O(h)$$

$$f''(x_i) = \frac{-f(x_{i+3}) + 4f(x_{i+2}) - 5f(x_{i+1}) + 2f(x_i)}{h^2} \qquad O(h^2)$$

Forward Finite-Divided Differences

Third Derivative

Error

$$f'''(x_i) = \frac{f(x_{i+3}) - 3f(x_{i+2}) + 3f(x_{i+1}) - f(x_i)}{h^3} \qquad O(h)$$

$$f'''(x_i) = \frac{-3f(x_{i+4}) + 14f(x_{i+3}) - 24f(x_{i+2}) + 18f(x_{i+1}) - 5f(x_i)}{2h^3} \qquad O(h^2)$$

Fourth Derivative

$$f^{(4)}(x_i) = \frac{f(x_{i+4}) - 4f(x_{i+3}) + 6f(x_{i+2}) - 4f(x_{i+1}) + f(x_i)}{h^4} \qquad O(h)$$

$$f^{(4)}(x_i) = \frac{-2f(x_{i+5}) + 11f(x_{i+4}) - 24f(x_{i+3}) + 26f(x_{i+2}) - 14f(x_{i+1}) + 3f(x_i)}{h^4} \qquad O(h^2)$$

Backward Finite-Divided Differences

First Derivative — **Error**

$$f'(x_i) = \frac{f(x_i) - f(x_{i-1})}{h} \qquad O(h)$$

$$f'(x_i) = \frac{3f(x_i) - 4f(x_{i-1}) + f(x_{i-2})}{2h} \qquad O(h^2)$$

Second Derivative

$$f''(x_i) = \frac{f(x_i) - 2f(x_{i-1}) + f(x_{i-2})}{h^2} \qquad O(h)$$

$$f''(x_i) = \frac{2f(x_i) - 5f(x_{i-1}) + 4f(x_{i-2}) - f(x_{i-3})}{h^2} \qquad O(h^2)$$

Backward Finite-Divided Differences

Third Derivative

Error

$$f'''(x_i) = \frac{f(x_i) - 3f(x_{i-1}) + 3f(x_{i-2}) - f(x_{i-3})}{h^3} \qquad O(h)$$

$$f'''(x_i) = \frac{5f(x_i) - 18f(x_{i-1}) + 24f(x_{i-2}) - 14f(x_{i-3}) + 3f(x_{i-4})}{2h^3} \qquad O(h^2)$$

Fourth Derivative

$$f^{(4)}(x_i) = \frac{f(x_i) - 4f(x_{i-1}) + 6f(x_{i-2}) - 4f(x_{i-3}) + f(x_{i-4})}{h^4} \qquad O(h)$$

$$f^{(4)}(x_i) = \frac{3f(x_i) - 14f(x_{i-1}) + 26f(x_{i-2}) - 24f(x_{i-3}) + 11f(x_{i-4}) - 2f(x_{i-5})}{h^4} \qquad O(h^2)$$